

CMP-22 - Event and Messaging Platform

Provide durable commands, queries, events, envelopes, ordering scopes, deduplication, outbox, thread-delivery, and delivery semantics.

Authority and state ownership

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 22, 23. Authoritative state: Event Journal, message envelopes, deduplication records, delivery and subscription state, Thread Delivery Records, Transactional Outbox publication state. Explicitly non-owned: domain authoritative state, command Permission evaluation, global ordering, business process ownership.

Critical invariant

Commands request possible change, Events record completed facts, and Queries do not intentionally mutate authoritative state.

Logical interfaces

CMP-22-IF-01 / Command / Publish Command

Command `Publish Command` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-02 / Command / Append Event

Command `Append Event` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-03 / Command / Acknowledge Delivery

Command `Acknowledge Delivery` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-04 / Command / Quarantine Message

Command `Quarantine Message` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-05 / Query / Query Event Stream

Query `Query Event Stream` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-06 / Query / Get Delivery State

Query `Get Delivery State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-07 / Query / Trace Causation

Query `Trace Causation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-08 / Event / Message Accepted

Event `Message Accepted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-09 / Event / Event Published

Event `Event Published` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-10 / Event / Delivery Failed

Event `Delivery Failed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-22-IF-11 / Event / Message Quarantined

Event `Message Quarantined` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Created > Accepted > Delivered > Acknowledged or Failed or Expired. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: duplicate delivery, causation loss, poison message, ordering overclaim, outbox divergence. Recovery proves identity, state, authority, provenance, dependencies, and

invariants before mutation resumes.

Conformance tests

CMP-22-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-22-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-22-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-22-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-22-TST-005 - Critical invariant. Commands request possible change, Events record completed facts, and Queries do not intentionally mutate authoritative state. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-22-TST-006 - Failure containment. Inject duplicate delivery, causation loss, poison message, ordering overclaim, outbox divergence. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-22-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-22-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-22-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-22-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.